NodeJS

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03 EventEmitter

04 Registering Callbacks

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10 Read Data from a file using streams

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Agenda

11 Write Data to a file using streams

12 Stream Events

13 Process Stream Events

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HTTP

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What is HTTP?

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What is HTTP?

HTTP stands for Hypertext Transfer Protocol. It is a hypertext protocol that is used on the internet

The communication between clients and servers on the web is done on the internet using the HTTP protocol

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HTTP Headers

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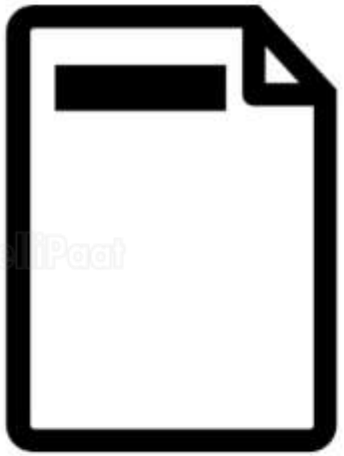
HTTP Headers

HTTP headers are used to enable the client and the server to pass additional information with an HTTP request or response.

An HTTP header can be thought of as a key-value pair which consists of a case-insensitive key followed by a colon (:), then by its value.



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HTTP Response Status Codes

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HTTP Response Status Codes

HTTP Status Codes are 3 digit numeric codes that are sent by the server to the client within the response. They indicate whether a response was completed successfully

The first digit of the response code define the class of response

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HTTP Response Classes

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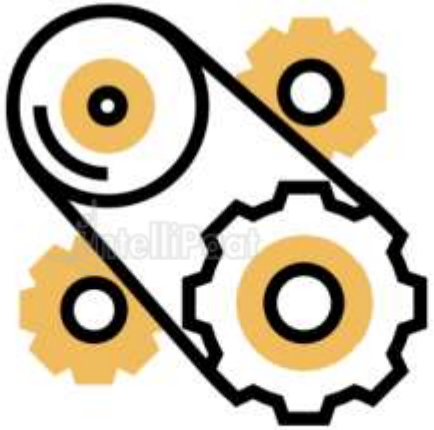
HTTP Response Classes

There are 5 HTTP Response Classes

- 1xx
- 2xx
- 3xx
- 4xx
- 5xx

Request has been received and the process is continuing



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HTTP Response Classes

There are 5 HTTP Response Classes

- 1xx
- 2xx
- 3xx
- 4xx
- 5xx

Action was successfully received, understood, and accepted

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HTTP Response Classes

There are 5 HTTP Response Classes

1xx

2xx

3xx

4xx

5xx

Further action must be taken in order to complete the request

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HTTP Response Classes

There are 5 HTTP Response Classes

1xx

2xx

3xx

4xx

5xx

The request contains incorrect syntax or cannot be fulfilled

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HTTP Response Classes

There are 5 HTTP Response Classes

- 1xx
- 2xx
- 3xx
- 4xx
- 5xx

The server failed to fulfill an apparently valid request

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Node HTTP Module

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Node HTTP Module

Node.js has a built-in module called HTTP, which allows Node.js to transfer data over the Hyper Text Transfer Protocol (HTTP)

The http module can be used to create an http server that can listen to requests on a specified port and respond to those requests

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Demo: Create an HTTP Server

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HTTP Form Data

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HTTP Form Data

Using the http protocol users can also send data submitted via a from

This form might be used to submit login credentials, a complaint and so on. This form data is sent to the server inside the request and can be parsed by the server to extract the user submitted data

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HTTP Form Data Request Type

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HTTP Form Data Request Type

There are 2 types of requests you can use to submit a form using HTTP protocol

GET

POST

A GET request gets the data from the form encodes it in the URL as query string parameters and sends it to the server. Data sent using GET request is visible in the URL

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HTTP Form Data Request Type

There are 2 types of requests you can use to submit a form using HTTP protocol

GET

POST

A POST request gets the data from the form encodes it sends it to the server in the request. Data sent using POST request is not visible in the URL

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Demo: Process Form Data

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Query string

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Node Query string Module

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Processing Form Data using HTTP Module

We can process the from data submitted to our server using the HTTP module

We do this by registering an event handler on the response object for the event 'data', and when we hit the 'end' request we can now process the from data

```
http://localhost:3000/form?username=anirudh&password=sample
```

URL

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Processing Form Data using HTTP Module

Query strings are used to pass data from client to server by encoding data in the URL

Query strings can be thought of as key-values pairs that are encoded in the url

```
http://localhost:3000/form?username=anirudh&password=sample
```

Query String

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Processing Form Data using HTTP Module

Query strings are appended at the end of the of the URL as key value pairs in the format 'key=value'

Query strings in URL begin after the URL and a question mark followed by key=value and multiple key-values are separated by the ampersand sign '&

http://localhost:3000/form?username=anirudh&password=sample

Key = username

value = anirudh

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Querystring module

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Querystring module

Node js comes with an in-built querystring module that allows you to deal with querystrings inside your node application

The querystring module contains a number of in-built methods that deal with encoding and decoding urls

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Querystring methods

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HTTP Response Classes

There are six important querystring methods

- encode
- decode
- escape
- parse
- stringify
- unescape

encode method produces a URL query string from a given object by iterating through the object

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HTTP Response Classes

There are six important querystring methods

- encode
- decode
- escape
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- stringify
- unescape

decode method parses a URL query string into a collection of key and value pairs

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HTTP Response Classes

There are six important querystring methods

- encode
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escape method performs URL encoding on the given string. It is used internally by stringify and encode methods

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HTTP Response Classes

There are six important querystring methods

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It is just an alias for the decode method of the querystring module

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HTTP Response Classes

There are six important querystring methods

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It is just an alias for the encode method of the querystring module

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HTTP Response Classes

There are six important querystring methods

- encode
- decode
- escape
- parse
- stringify
- unescape

unescape method performs URL decoding on the given string. It is used internally by parse and decode methods

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Encode Querystring

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Decode Querystring

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Quiz

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
D	readFile

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Quiz

1. Which method is used to read file synchronously?

A	readFileSynchronously
B	readFileSync
C	readFS
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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

2. Which method is used to read file asynchronously?

A	readFileAsynchronously
B	readFileAsync
C	readFAS
D	readFile

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

3. Which constant is NodeJS is used to refer to absolute path of the current file?

A	__filename
B	__dirname
C	__absoule_file_path
D	__abs_filename

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

4. Which constant is NodeJS is used to refer to absolute path of the current directory?

A	__filename
B	__dirname
C	__absoule_dir_path
D	__abs_directory

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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Quiz

5. Which property of the filestats object can be used to determine the time the file was last accessed?

A	access_timestamp
B	birthname
C	atime
D	uid

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India: +91-7847955955

US: 1-800-216-8930 (TOLL FREE)

support@intellipaat.com

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